

Artificial Intelligence Notes

Chapter 1: Introduction

Prepared from: Khemani, 'A First Course in Artificial Intelligence'

History and Applications of AI

Artificial Intelligence traces its origins to the 1956 Dartmouth conference. Over time it has progressed from symbolic reasoning and expert systems to machine learning and deep learning. Applications span medicine, robotics, natural language processing, computer vision, games, and finance. Artificial Intelligence traces its origins to the 1956 Dartmouth conference. Over time it has progressed from symbolic reasoning and expert systems to machine learning and deep learning. Applications span medicine, robotics, natural language processing, computer vision, games, and finance. Artificial Intelligence traces its origins to the 1956 Dartmouth conference. Over time it has progressed from symbolic reasoning and expert systems to machine learning and deep learning. Applications span medicine, robotics, natural language processing, computer vision, games, and finance. Artificial Intelligence traces its origins to the 1956 Dartmouth conference. Over time it has progressed from symbolic reasoning and expert systems to machine learning and deep learning. Applications span medicine, robotics, natural language processing, computer vision, games, and finance. Artificial Intelligence traces its origins to the 1956 Dartmouth conference. Over time it has progressed from symbolic reasoning and expert systems to machine learning and deep learning. Applications span medicine, robotics, natural language processing, computer vision, games, and finance.

AI Techniques

AI techniques include search algorithms, heuristic methods, knowledge representation, reasoning, and learning. These techniques reduce problem complexity and enable intelligent behavior in dynamic environments. AI techniques include search algorithms, heuristic methods, knowledge representation, reasoning, and learning. These techniques reduce problem complexity and enable intelligent behavior in dynamic environments. AI techniques include search algorithms, heuristic methods, knowledge representation, reasoning, and learning. These techniques reduce problem complexity and enable intelligent behavior in dynamic environments. AI techniques include search algorithms, heuristic methods, knowledge representation, reasoning, and learning. These techniques reduce problem complexity and enable intelligent behavior in dynamic environments. AI techniques include search algorithms, heuristic methods, knowledge representation, reasoning, and learning. These techniques reduce problem complexity and enable intelligent behavior in dynamic environments. AI techniques include search algorithms, heuristic methods, knowledge representation, reasoning, and learning. These techniques reduce problem complexity and enable intelligent behavior in dynamic environments. AI techniques include search algorithms, heuristic methods, knowledge representation, reasoning, and learning. These techniques reduce problem complexity and enable intelligent behavior in dynamic environments.

Types of AI

AI can be categorized into Narrow AI, General AI, and Super AI. It can also be described as reactive, limited memory, theory of mind, and self-aware systems. AI can be categorized into Narrow AI, General AI, and Super AI. It can also be described as reactive, limited memory, theory of mind, and self-aware systems. AI can be categorized into Narrow AI, General AI, and Super AI. It can also be described as reactive, limited memory, theory of mind, and self-aware systems.

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Intelligent Agent: Types

[illegible]

Intelligent Agent: Environment

[illegible]

Solving Problems by Searching

Toy problems

Toy problems are simplified examples like the 8-puzzle or Tic-Tac-Toe. They illustrate search principles and are easier to solve. Toy problems are simplified examples like the 8-puzzle or Tic-Tac-Toe. They illustrate search principles and are easier to solve. Toy problems are simplified examples like the 8-puzzle or Tic-Tac-Toe. They illustrate search principles and are easier to solve. Toy problems are simplified examples like the 8-puzzle or Tic-Tac-Toe. They illustrate search principles and are easier to solve. Toy problems are simplified examples like the 8-puzzle or Tic-Tac-Toe. They illustrate search principles and are easier to solve. Toy problems are simplified examples like the 8-puzzle or Tic-Tac-Toe. They illustrate search principles and are easier to solve. Toy problems are simplified examples like the 8-puzzle or Tic-Tac-Toe. They illustrate search principles and are easier to solve.

Real-world problems

Real-world problems include navigation, scheduling, and optimization. These are large-scale, uncertain, and computationally challenging. Real-world problems include navigation, scheduling, and optimization. These are large-scale, uncertain, and computationally challenging. Real-world problems include navigation, scheduling, and optimization. These are large-scale, uncertain, and computationally challenging. Real-world problems include navigation, scheduling, and optimization. These are large-scale, uncertain, and computationally challenging. Real-world problems include navigation, scheduling, and optimization. These are large-scale, uncertain, and computationally challenging. Real-world problems include navigation, scheduling, and optimization. These are large-scale, uncertain, and computationally challenging. Real-world problems include navigation, scheduling, and optimization. These are large-scale, uncertain, and computationally challenging.

8 Puzzle Game

The 8-puzzle is solved using algorithms like BFS, DFS, and A*, with heuristics such as misplaced tiles and Manhattan distance. The 8-puzzle is solved using algorithms like BFS, DFS, and A*, with heuristics such as misplaced tiles and Manhattan distance. The 8-puzzle is solved using algorithms like BFS, DFS, and A*, with heuristics such as misplaced tiles and Manhattan distance. The 8-puzzle is solved using algorithms like BFS, DFS, and A*, with heuristics such as misplaced tiles and Manhattan distance. The 8-puzzle is solved using algorithms like BFS, DFS, and A*, with heuristics such as misplaced tiles and Manhattan distance. The 8-puzzle is solved using algorithms like BFS, DFS, and A*, with heuristics such as misplaced tiles and Manhattan distance. The 8-puzzle is solved using algorithms like BFS, DFS, and A*, with heuristics such as misplaced tiles and Manhattan distance.

Chessboard problem

Chess demonstrates large branching factors. Minimax with alpha-beta pruning is applied to reduce complexity. Chess demonstrates large branching factors. Minimax with alpha-beta

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Tic-tac-toe

Tic-Tac-Toe shows decision tree strategies and can be exhaustively solved for win/loss/draw. Tic-Tac-Toe shows decision tree strategies and can be exhaustively solved for win/loss/draw. Tic-Tac-Toe shows decision tree strategies and can be exhaustively solved for win/loss/draw. Tic-Tac-Toe shows decision tree strategies and can be exhaustively solved for win/loss/draw. Tic-Tac-Toe shows decision tree strategies and can be exhaustively solved for win/loss/draw. Tic-Tac-Toe shows decision tree strategies and can be exhaustively solved for win/loss/draw.

Water-jug problem

The water jug problem illustrates state space representation, operators, and goal testing. The water jug problem illustrates state space representation, operators, and goal testing. The water jug problem illustrates state space representation, operators, and goal testing. The water jug problem illustrates state space representation, operators, and goal testing. The water jug problem illustrates state space representation, operators, and goal testing.

Travelling Salesman Problem

TSP is an NP-hard optimization problem solved with heuristics and approximation algorithms. TSP is an NP-hard optimization problem solved with heuristics and approximation algorithms. TSP is an NP-hard optimization problem solved with heuristics and approximation algorithms. TSP is an NP-hard optimization problem solved with heuristics and approximation algorithms. TSP is an NP-hard optimization problem solved with heuristics and approximation algorithms.

Wumpus World Problem

Wumpus World illustrates reasoning under uncertainty and knowledge-based agents. Wumpus World illustrates reasoning under uncertainty and knowledge-based agents. Wumpus World illustrates reasoning under uncertainty and knowledge-based agents. Wumpus World illustrates reasoning under uncertainty and knowledge-based agents.

agents. Wumpus World illustrates reasoning under uncertainty and knowledge-based agents.

Key Points Summary

- AI evolved from symbolic reasoning to machine learning.
- Applications include NLP, robotics, vision, medicine.
- Techniques include search, reasoning, knowledge representation, learning.
- Agents are designed according to environment properties.
- Search problems highlight algorithmic strategies and heuristics.